Equality, rigidity and full-ellipsoid CHSH envelopes at fixed steering-ellipsoid centre

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Abstract

Milne, Jennings, Jevtic and Rudolph conjectured that a two-qubit state whose steering-ellipsoid centre has length c has CHSH value at most $\max\{2\sqrt{2(1-c)}, 2\}$ (arXiv:1404.3951, Conjecture 1). The Omega Institute's public repository recorded a claimed Lean proof of that conjecture on 14 October 2026, and its proof already contains the refined bound $\beta^2/4 \leq 1 + (1-r^2)(1-2c)$, where r is the length of Bob's Bloch vector; an attaining family with this value goes back to Zhang, Araiza Bravo, Lorenz and Chitambar (2020). We claim no priority for either. Building on them, we prove the complete equality classification at every fixed pair (c, r): for $0 < c < 1$ the extremizers form a single local-unitary class, a filtered maximally obese state with the filter condition $\|\boldsymbol{b}\|^2 = b_z$. We give a quantitative trace-norm rigidity theorem, including an estimate that preserves the whole centre vector with exponent $1/4$, and a separable rank-two family showing that this exponent cannot be improved uniformly in c. Beyond the centre alone, we determine the exact fixed-radius CHSH envelope for every axial steering ellipsoid, with all equality directions and a resulting exact CHSH-breaking criterion for axial qubit channels, and for every centred ellipsoid. We also correct a printed optimizing direction in the 2020 paper. A Lean 4 development checks, for every admissible (c, r), an actual complex density matrix with the prescribed centre and Bob length whose CHSH supremum over all unit Pauli settings is exactly $2\sqrt{1 + (1-2c)(1-r^2)}$, two further explicit witnesses, and the scalar spectral optimization behind the axial envelope; the classification, rigidity and general filtering steps are conventional proofs.

1 Introduction

For a two-qubit state, the steering ellipsoid of Alice is the set of Bloch vectors to which Bob can steer her by local measurements and postselection. Milne, Jennings, Jevtic and Rudolph [1] proved that, for states whose Bob marginal is maximally mixed, the CHSH value is at most $2\sqrt{2(1-c)}$, where c is the length of the ellipsoid's centre, and they conjectured (Conjecture 1 of [1]) that every two-qubit state obeys

$$\beta(\rho) \leq \max\{2\sqrt{2(1-c)}, 2\}. \tag{1}$$

Ducuara, Susa and Skrzypczyk [4] use centre bounds of this kind in their criterion for locally inaccessible hidden correlations.

Earlier public work and priority (note dated 3 November 2026). The Omega Institute's public repository `the-omega-institute/trureturing` records a claimed Lean proof of Conjecture 1 for every two-qubit density matrix, with supremal sharpness at every centre length; the dossier for its issue #16558 was committed on 14 October 2026 at 08:40:22 UTC [2]. Its proof combines the

inequalities $M \le 1 + 2\Delta(\rho)$ and $\Delta(\rho) \le (1 - r^2)(\frac{1}{2} - c)$ (notation below), so it proves the refined joint inequality

$$M(\rho) := \frac{\beta(\rho)^2}{4} \le H(c, r) := 1 + (1 - r^2)(1 - 2c), \tag{2}$$

and the dossier states this inequality explicitly as proved on the way. The same record proves Conjecture 2 of [1] (issue #14206, committed on 8 October 2026). Independently of both, Appendix B of Zhang, Araiza Bravo, Lorenz and Chitambar [3] computes the optimum $1 + \alpha^2(2p - 1)$ of CHSH over an amplitude-damping family; with $p = 1 - c$ and $\alpha^2 = 1 - r^2$ this is $H(c, r)$. Combining the two, the exact supremum of β at fixed (c, r) is $2\sqrt{H(c, r)}$. *We claim no priority for Conjecture 1, for the joint bound* (2)*, for its attaining value, or for the attaining family.* Section 3 reconstructs the bound only to fix notation and to extract the slack that the new results use. We have not compiled the Omega Lean proof and describe it as claimed; its mathematics was read and reconstructed.

What is new here. The following go beyond Conjecture 1 and the Omega record. The Omega dossier lists bounds involving the full steering ellipsoid as open and argues attainment only for the centre-only bound.

1. *Equality classification* (Theorem 5.1). For $0 < c < 1$ and $r < 1$, $M = H(c, r)$ holds exactly for one local-unitary class: a maximally obese canonical state filtered by $\sqrt{\mathbb{I} + \boldsymbol{b} \cdot \boldsymbol{\sigma}}$ with $\|\boldsymbol{b}\|^2 = b_z$. The necessity step is new; earlier work exhibits attaining states but does not exclude others. Equality in (1) is classified for every c (Corollary 5.4).

2. *Rigidity* (Section 6). Near-equality forces closeness, in trace norm, to an exact extremizer with the same c and r; a version preserves the whole centre vector. Uniformly in c, canonical states lie within $\min\{2, 9D_{\mathrm{pur}}^{1/4}\}$ of a maximally obese state with the same Alice marginal, and the exponent $1/4$ is optimal.

3. *Full-ellipsoid envelopes* (Section 7). Exact fixed-radius CHSH maxima, with all optimizing filter directions, for every translated sphere, every axial ellipsoid (centre on the symmetry axis, oblate or prolate) and every centred ellipsoid, and the closed optimization over the radius. For axial qubit channels $x \mapsto ce_z + \mathrm{diag}(s, s, t)x$ this gives an exact criterion for CHSH breaking.

4. *A printed optimizer* (Section 8). The optimizing direction displayed in Appendix B of [3] is incorrect; an exact example shows it does not attain the stated maximum. The maximal value and threshold of [3] are unaffected.

The literature comparison is bounded; it does not establish worldwide novelty.

Verification. A standalone Lean 4.19.0 project (Section 9) proves, without assuming the Horodecki formula, that an explicit complex 4×4 density matrix with centre $(0, 0, c)$ and Bob length r has CHSH supremum exactly $2\sqrt{H(c, r)}$ for every admissible (c, r), and it checks the axial witness of Section 7.3, the sharpness family of Corollary 6.9 and the scalar spectral optimization of Theorem 7.2. The general filtering correspondence, the equality classification and the rigidity theorems are conventional proofs, supported by exact symbolic identities and seeded numerical falsification searches.

2 Definitions and conventions

A two-qubit density matrix has the Bloch representation

$$\rho = \frac{1}{4}\Big(\mathbb{I}\otimes\mathbb{I} + \boldsymbol{a}\cdot\boldsymbol{\sigma}\otimes\mathbb{I} + \mathbb{I}\otimes\boldsymbol{b}\cdot\boldsymbol{\sigma} + \sum_{i,j=1}^{3} T_{ij}\sigma_i\otimes\sigma_j\Big), \tag{3}$$

with Alice the first factor. Put $r = \|\boldsymbol{b}\|$. Following [1], four unit vectors define

$$B = (\alpha\cdot\boldsymbol{\sigma})\otimes((\beta+\beta')\cdot\boldsymbol{\sigma}) + (\alpha'\cdot\boldsymbol{\sigma})\otimes((\beta-\beta')\cdot\boldsymbol{\sigma}), \qquad \beta(\rho) = \max_{B} |\operatorname{tr}(\rho B)|.$$

Constant observables $\pm\mathbb{I}$ are not allowed, so β is not floored at two. By the Horodecki theorem [5],

$$M(\rho) = \frac{\beta(\rho)^2}{4} = \lambda_1(TT^{\mathsf{T}}) + \lambda_2(TT^{\mathsf{T}}) = \max_{u\perp v,\ \|u\|=\|v\|=1} \big(\big\|T^{\mathsf{T}}u\big\|^2 + \big\|T^{\mathsf{T}}v\big\|^2\big). \tag{4}$$

For $r < 1$, Alice's steering ellipsoid has centre and canonical form

$$\boldsymbol{c} = \frac{\boldsymbol{a} - T\boldsymbol{b}}{1-r^2}, \quad c = \|\boldsymbol{c}\|, \qquad \sigma = (\mathbb{I}\otimes(2\rho_B)^{-1/2})\rho(\mathbb{I}\otimes(2\rho_B)^{-1/2}), \quad \sigma_B = \mathbb{I}/2. \tag{5}$$

The canonical state σ has Alice Bloch vector $\boldsymbol{c}$ and correlation matrix C; the shape matrix is $Q = CC^{\mathsf{T}}$, whose eigenvalues are the squared semiaxes. For $r = 1$ positivity forces $\rho = \rho_A\otimes\rho_B$, the ellipsoid is the point $\{\boldsymbol{a}\}$ and we set $\boldsymbol{c} = \boldsymbol{a}$, the convention also used in [2].

The centre formula. For a Bob effect proportional to $\mathbb{I} + e\cdot\boldsymbol{\sigma}$, $\|e\| \le 1$, Alice's conditional Bloch vector is $(\boldsymbol{a} + Te)/(1 + \boldsymbol{b}\cdot e)$. An invertible positive Bob filter F maps effects bijectively, up to positive rescaling, by $E \mapsto FEF$, so whitening preserves the steering set. Put $s = \sqrt{1-r^2}$, $W = s\mathbb{I} + \boldsymbol{b}\boldsymbol{b}^{\mathsf{T}}/(1+s)$ and $F = \sqrt{\mathbb{I} + \boldsymbol{b}\cdot\boldsymbol{\sigma}}$. Pauli multiplication gives $F(x\cdot\boldsymbol{\sigma})F = (\boldsymbol{b}\cdot x)\mathbb{I} + (Wx)\cdot\boldsymbol{\sigma}$, $W\boldsymbol{b} = \boldsymbol{b}$ and $W^2 = s^2\mathbb{I} + \boldsymbol{b}\boldsymbol{b}^{\mathsf{T}}$. If the canonical state has Alice vector a_0 and correlation matrix C, undoing the whitening gives $\boldsymbol{a} = a_0 + C\boldsymbol{b}$, $T = a_0\boldsymbol{b}^{\mathsf{T}} + CW$ and $\boldsymbol{a} - T\boldsymbol{b} = (1-r^2)a_0$. The canonical steering set is $\{a_0 + Ce : \|e\| \le 1\}$, with centre a_0 and shape matrix CC^{T}; hence $a_0 = \boldsymbol{c}$. The same identities give the Gram formula

$$\boldsymbol{a} = \boldsymbol{c} + C\boldsymbol{b}, \qquad TT^{\mathsf{T}} = \boldsymbol{a}\boldsymbol{a}^{\mathsf{T}} + (1-r^2)(Q - \boldsymbol{c}\boldsymbol{c}^{\mathsf{T}}), \tag{6}$$

valid for a state $\rho = (\mathbb{I}\otimes F_{\boldsymbol{b}})\sigma(\mathbb{I}\otimes F_{\boldsymbol{b}})$ obtained from a canonical state by the filter $F_{\boldsymbol{b}} = \sqrt{\mathbb{I} + \boldsymbol{b}\cdot\boldsymbol{\sigma}}$. Every state with $r < 1$ has this form up to a local unitary on Bob, since $\rho = (\mathbb{I}\otimes\sqrt{2\rho_B})\sigma(\mathbb{I}\otimes\sqrt{2\rho_B})$.

3 The joint bound (prior work)

This section reconstructs (2) along the route of the Omega proof [2]; nothing in it is claimed as new. Define $\Delta(X) = \operatorname{tr}(X^2) - \operatorname{tr}((\operatorname{tr}_B X)^2)$.

Lemma 3.1 (CHSH–purity bound)**.** *For every two-qubit density matrix, $M(\rho) \le 1 + 2\Delta(\rho)$.*

Proof. Let u, v, n be an orthonormal frame, $K_n = \sigma_y\otimes((n\cdot\boldsymbol{\sigma})\sigma_y)$ and $\rho_n = K_n\rho^{\mathsf{T}}K_n^{\dagger} \succeq 0$ (full transpose). The map $\rho \mapsto \rho_n$ sends $(\boldsymbol{a}, \boldsymbol{b}, T)$ to $(-\boldsymbol{a}, \boldsymbol{b} - 2(\boldsymbol{b}\cdot n)n, T(2nn^{\mathsf{T}} - \mathbb{I}))$, so Pauli orthogonality gives $4\operatorname{tr}(\rho\rho_n) = 1 - \|\boldsymbol{a}\|^2 + \|\boldsymbol{b}\|^2 - 2(\boldsymbol{b}\cdot n)^2 + 2\|Tn\|^2 - \|T\|_F^2$. With $4\Delta = \|T\|_F^2 + \|\boldsymbol{b}\|^2 - \|\boldsymbol{a}\|^2 - 1$,

$$\|Tu\|^2 + \|Tv\|^2 + (\boldsymbol{b}\cdot n)^2 + 2\operatorname{tr}(\rho\rho_n) = 1 + 2\Delta(\rho). \tag{7}$$

The last two terms on the left are nonnegative; maximize over the plane. □

Lemma 3.2 (Filter covariance)**.** *For every 4×4 matrix X and every complex 2×2 matrix F, $\Delta((\mathbb{I} \otimes F)X(\mathbb{I} \otimes F)^\dagger) = |\det F|^2 \Delta(X)$.*

Proof. For 2×2 matrices, $\operatorname{tr}(AB) - \operatorname{tr} A \operatorname{tr} B = \det A + \det B - \det(A + B)$, and simultaneous congruence by F multiplies the right side by $|\det F|^2$. Apply this to the Bob blocks of X and sum over Alice indices. □

Lemma 3.3 (Canonical caps)**.** *If $\sigma_B = \mathbb{I}/2$ and Alice's Bloch length is c, then $\lambda_{\max}(\sigma) \le L := 1 - c/2$, $\operatorname{tr} \sigma^2 \le L^2 + (1 - L)^2 = 1 - c + c^2/2$ and $\Delta(\sigma) \le 1/2 - c$.*

Proof. Let λ be the largest eigenvalue with unit eigenvector ψ and $\tau = \sigma - \lambda|\psi\rangle\langle\psi| \succeq 0$. Bob's marginal gives $\lambda\psi_B \preceq \mathbb{I}/2$; the marginals of a pure state have equal spectra, so $\lambda\psi_A \preceq \mathbb{I}/2$. For a top eigenvector e_+ of σ_A, $1 - \lambda = \operatorname{tr} \tau_A \ge \langle e_+|\tau_A|e_+\rangle = \frac{1+c}{2} - \lambda\langle e_+|\psi_A|e_+\rangle \ge c/2$. If $\lambda \le 1/2$ then $\operatorname{tr} \sigma^2 \le \lambda \le 1/2 \le L^2 + (1 - L)^2$; otherwise $\operatorname{tr} \sigma^2 \le \lambda^2 + (1 - \lambda)^2$, increasing for $\lambda \ge 1/2$. Subtract Alice's purity $(1 + c^2)/2$. □

For $r < 1$ write $\rho = (\mathbb{I} \otimes F)\sigma(\mathbb{I} \otimes F)$ with $F = \sqrt{2\rho_B}$ and $|\det F|^2 = 1 - r^2$; the three lemmas give (2). If $r = 1$ the state is a product and $M = c^2 \le 1 = H$. Maximizing H over r gives (1).

4 Canonical equality and a shape-sensitive deficit

Define the canonical maximally obese state

$$\omega_c = |w_c\rangle\langle w_c| + \frac{c}{2}|01\rangle\langle 01|, \qquad |w_c\rangle = \frac{|00\rangle + \sqrt{1-c}|11\rangle}{\sqrt{2}}, \tag{8}$$

with $\boldsymbol{c} = ce_z$, $C = \operatorname{diag}(\sqrt{1-c}, -\sqrt{1-c}, 1-c)$ and $\operatorname{spec}(\omega_c) = \{1 - c/2, c/2, 0, 0\}$. It is locally unitarily equivalent to the maximally obese state of [1].

Theorem 4.1 (Canonical equality)**.** *Equality in the spectral cap or in the purity cap of Lemma 3.3 holds if and only if σ is a local-unitary image of ω_c.*

Proof. Let $0 < c \le 1$ and $\lambda = L$. Every inequality in the proof of the spectral cap is an equality, so $\tau_A = \frac{c}{2}|e_+\rangle\langle e_+|$ and $\lambda\psi_A = \frac{1}{2}|e_+\rangle\langle e_+| + \frac{1-c}{2}|e_-\rangle\langle e_-|$. A positive operator with a rank-one marginal is a product on that factor, so $\tau = |e_+\rangle\langle e_+| \otimes \tau_B$. In Schmidt bases $\sqrt{\lambda}|\psi\rangle = |e_+, f_0\rangle/\sqrt{2} + \sqrt{(1-c)/2}\,|e_-, f_1\rangle$, and Bob's marginal forces $\tau_B = \frac{c}{2}|f_1\rangle\langle f_1|$: this is (8) in local bases. At $c = 0$, $\lambda = 1$ gives a pure state with maximally mixed Bob marginal, a maximally entangled state. The converse is immediate. For $c < 1$ the purity cap exceeds $1/2$, so equality in it forces $\lambda > 1/2$ and, by strict monotonicity of $x^2 + (1 - x)^2$, $\lambda = L$. At $c = 1$ every canonical state is $|e_+\rangle\langle e_+| \otimes \mathbb{I}/2$. □

Define the nonnegative shape deficit

$$D_{\text{pur}} = 1 - c + \frac{c^2}{2} - \operatorname{tr} \sigma^2 = \frac{3 - 4c + c^2 - \operatorname{tr} Q}{4}, \tag{9}$$

using $4 \operatorname{tr} \sigma^2 = 1 + c^2 + \operatorname{tr} Q$. Theorem 4.1 says $D_{\text{pur}} = 0$ exactly on the obese orbit. Keeping this slack in the chain of Section 3 gives

$$M(\rho) \le H(c, r) - 2(1 - r^2)D_{\text{pur}} = 1 + \frac{1 - r^2}{2}(\operatorname{tr} Q - c^2 - 1). \tag{10}$$

This is an immediate refinement of the prior chain, not a sharp bound for a general ellipsoid, since orientation is discarded; it penalizes every canonical state that is not maximally obese.

5 Equality in the joint bound

Theorem 5.1 (Equality classification)**.** *For $0 \le c \le 1$ and $0 \le r < 1$,*

$$\max_{\rho:\ \|\boldsymbol{c}\|=c,\ \|\boldsymbol{b}\|=r} \beta(\rho) = 2\sqrt{1+(1-r^2)(1-2c)}. \tag{11}$$

For $0 < c < 1$ the attaining states are, up to local unitaries, exactly

$$\rho_{c,\boldsymbol{b}} = (\mathbb{I}\otimes F_{\boldsymbol{b}})\omega_c(\mathbb{I}\otimes F_{\boldsymbol{b}}), \qquad F_{\boldsymbol{b}} = \sqrt{\mathbb{I}+\boldsymbol{b}\cdot\boldsymbol{\sigma}}, \qquad \|\boldsymbol{b}\|^2 = b_z < 1, \tag{12}$$

for instance $\boldsymbol{b} = (r\sqrt{1-r^2}, 0, r^2)$. All transverse azimuths are equivalent under local unitaries fixing ω_c, so each interior pair (c, r) has one local-unitary class of extremizers. At $c = 0$ the extremizers are the pure states with Bob length r; at $c = 1$ every state is a product with pure Alice marginal and attains $2r$.

The value (11) is prior work (Section 1); the classification is the new content.

Proof. The upper bound is (2). If equality holds with $r < 1$, every inequality in $M \le 1 + 2\Delta(\rho) = 1 + 2(1-r^2)\Delta(\sigma) \le H$ is an equality, so $D_{\mathrm{pur}} = 0$, and Theorem 4.1 reduces every extremizer to a filtered ω_c (the unitary freedom in F is absorbed into local unitaries). It remains to find the filters that attain the cap. Let $R = \|\boldsymbol{b}\|^2$, $p = 1-c$, and rotate so that $\boldsymbol{b} = (\sqrt{u}, 0, z)$, $u \ge 0$, $R = u + z^2 < 1$. By (6), with $q = p(1-R)$, $k = (1-2c)(1-R)$ and $h = 1 + k - q = p + cR$, the y direction is an eigenvector of TT^{T} with eigenvalue q, and the xz block is

$$B = \begin{pmatrix} p(1-z^2) & \sqrt{pu}(c+pz) \\ \sqrt{pu}(c+pz) & k + (c+pz)^2 \end{pmatrix}. \tag{13}$$

Its eigenvalues $\lambda_- \le \lambda_+$ satisfy $\lambda_- \le q \le \lambda_+$, because $\det(q\mathbb{I} - B) = -cp\,u(1-R) \le 0$; hence $M = q + \lambda_+$. Moreover

$$(h\mathbb{I}-B)_{11} = cu + z^2, \qquad (h\mathbb{I}-B)_{22} = p\{u + c(z-1)^2\}, \qquad \det(h\mathbb{I}-B) = cp(R-z)^2, \tag{14}$$

so $h\mathbb{I} - B \succeq 0$ and $M \le q + h = H$. For $0 < c < 1$ equality holds exactly when $\det(h\mathbb{I} - B) = 0$, that is $R = z$. This gives both directions of the classification; the displayed $\boldsymbol{b}$ has $R = r^2 = z$. The filter is invertible and positive and yields trace one because $(\omega_c)_B = \mathbb{I}/2$; scaled by $1/\sqrt{1+r}$ it is a valid Kraus operator. At $c = 0$, $D_{\mathrm{pur}} = 0$ forces a maximally entangled canonical state, and invertible filtering gives exactly the entangled pure states with $r < 1$, with $M = 2 - r^2$. At $c = 1$ a pure Alice marginal forces a product and $M = r^2$. □

For $0 < c < 1$ and $r < 1$ the classification reads, without coordinates,

$$M = H \iff D_{\mathrm{pur}} = 0 \text{ and } \boldsymbol{c}\cdot(\boldsymbol{a}-\boldsymbol{c}) = c(1-c)r^2,$$

since $\boldsymbol{a} - \boldsymbol{c} = C\boldsymbol{b}$ and in the obese frame the left side is $c(1-c)b_z$.

Proposition 5.2 (Explicit extremizer)**.** *For $0 \le c \le 1$ and $0 \le r < 1$ put $p = 1-c$, $s = \sqrt{1-r^2}$, $a = \sqrt{p}\,s$,*

$$v_0 = \tfrac{1}{2}(1+r, -(1-r), a, a)^{\mathsf{T}}, \qquad v_1 = \tfrac{\sqrt{c}\,s}{2}(1,1,0,0)^{\mathsf{T}}, \qquad \rho^{\sharp}_{c,r} = v_0 v_0^{\mathsf{T}} + v_1 v_1^{\mathsf{T}}.$$

Then $\rho^{\sharp}_{c,r}$ is a density matrix with $\boldsymbol{b} = (0,0,r)$, $\boldsymbol{a} = (ra, 0, c+pr^2)$, $\boldsymbol{c} = (0,0,c)$,

$$T = \begin{pmatrix} ra & 0 & a \\ 0 & -a & 0 \\ -p(1-r^2) & 0 & r \end{pmatrix}, \qquad \det(x\mathbb{I} - TT^{\mathsf{T}}) = (x-h)(x-q)(x-qh),$$

with $h = p + cr^2 \geq q = p(1-r^2) \geq qh$; consequently $M = h + q = H(c,r)$.

Proof. Positivity is manifest and $\|v_0\|^2 + \|v_1\|^2 = 1$. Partial traces give $\rho_B = \mathrm{diag}((1+r)/2, (1-r)/2)$ and the stated $\boldsymbol{a}$; Pauli traces give T; $\boldsymbol{a} - T\boldsymbol{b} = (0, 0, c(1-r^2))$ gives the centre. The characteristic polynomial and ordering follow from $a^2 = q$, $h - q = r^2$ and $q - qh = qc(1-r^2)$. Whitening gives $C = \mathrm{diag}(\sqrt{p}, -\sqrt{p}, p)R_y$ with a rotation R_y, so $Q = \mathrm{diag}(p, p, p^2)$. □

This is a coordinate realization of the attaining family discussed above, not a new family. Its Bell supremum is the formal endpoint of Section 9.

Remark 5.3 (Pure Bob marginal). At $r = 1$ the constrained maximum is $2c$, not the limit 2 of (11): the state is a product and its centre is $\boldsymbol{a}$. The centre can jump when an invertible filter becomes rank one.

Corollary 5.4 (Equality in Conjecture 1)**.** *Up to local unitaries, equality in* (1) *holds exactly as follows: for $0 \leq c < 1/2$, only for ω_c; for $c = 1/2$, exactly for the states $\rho_{1/2,\boldsymbol{b}}$ with $R = b_z < 1$, whose filter vectors form the sphere $b_x^2 + b_y^2 + (b_z - 1/2)^2 = 1/4$ without its north pole; for $1/2 < c < 1$, never (the supremum two is not attained); for $c = 1$, exactly for pure product states.*

Proof. For $c < 1/2$, $H = 2(1-c) - r^2(1-2c)$ forces $r = 0$. For $c = 1/2$, $H = 1$ for every $r < 1$, and Theorem 5.1 applies. For $1/2 < c < 1$, $H < 1$ when $r < 1$, and a pure Bob marginal gives $M = c^2 < 1$. For $c = 1$, $M = r^2$ requires $r = 1$. □

The Omega dossier argues, without formalizing it, that (1) is attained exactly for $c \in [0, 1/2] \cup \{1\}$; the corollary adds the classification of the attaining states. At $c = 1/2$ equality does not force a maximally mixed Bob marginal: the matrix

$$\rho_* = \frac{1}{32}\begin{pmatrix} 21 & -3 & 3\sqrt{6} & 3\sqrt{6} \\ -3 & 5 & -\sqrt{6} & -\sqrt{6} \\ 3\sqrt{6} & -\sqrt{6} & 3 & 3 \\ 3\sqrt{6} & -\sqrt{6} & 3 & 3 \end{pmatrix} = v_0 v_0^{\mathsf{T}} + v_1 v_1^{\mathsf{T}}, \quad v_0 = \left(\tfrac{3}{4}, -\tfrac{1}{4}, \tfrac{\sqrt{6}}{8}, \tfrac{\sqrt{6}}{8}\right)^{\mathsf{T}},\ v_1 = \left(\tfrac{\sqrt{6}}{8}, \tfrac{\sqrt{6}}{8}, 0, 0\right)^{\mathsf{T}},$$

has $\rho_B = \mathrm{diag}(3/4, 1/4)$, $\boldsymbol{c} = (0, 0, 1/2)$, $\mathrm{spec}(TT^{\mathsf{T}}) = \{5/8, 3/8, 15/64\}$ and hence $\beta(\rho_*) = 2$.

6 Quantitative rigidity

The norm $\|\cdot\|_1$ is the full trace norm (twice the trace distance).

Lemma 6.1 (Spectral deficit)**.** *Let $0 \leq c < 1$, $p = 1 - c$, $L = 1 - c/2$ and $\delta = L - \lambda_{\max}(\sigma)$ for a canonical σ. Then $0 \leq \delta \leq D_{\mathrm{pur}}/p$.*

Proof. With $m = \lambda_{\max}$ we have $\mathrm{tr}\,\sigma^2 \leq c/2 + pm$: if $m \leq 1/2$ use $\mathrm{tr}\,\sigma^2 \leq m$; if $m \geq 1/2$ use $\mathrm{tr}\,\sigma^2 \leq m^2 + (1-m)^2$ and $c/2 + pm - m^2 - (1-m)^2 = (L-m)(2m-1) \geq 0$. Subtract from $L^2 + (1-L)^2$ and use $L^2 + (1-L)^2 - pL = c/2$. □

Theorem 6.2 (Trace-norm rigidity)**.** *If $0 \le c < 1$ and $D_{\mathrm{pur}} \le (1-c)^2/4$, some local-unitary image ω of ω_c satisfies*

$$\|\sigma - \omega\|_1 \le 2\sqrt{\frac{D_{\mathrm{pur}}}{1-c}} + \frac{2\sqrt{D_{\mathrm{pur}}}}{1-c} + \frac{D_{\mathrm{pur}}}{1-c} \le \frac{9\sqrt{D_{\mathrm{pur}}}}{2(1-c)}. \tag{15}$$

The target has the same centre length; the constants are not claimed sharp.

Proof. Let $\lambda = L - \delta$ be the largest eigenvalue, $\sigma = |v\rangle\langle v| + \tau$ with $v = \sqrt{\lambda}\psi$, $\tau \succeq 0$, $\tau v = 0$, $d = c/2$ and $t = \operatorname{tr}\tau = d + \delta$. In Schmidt bases $v = \sqrt{s_+}|e_0 f_0\rangle + \sqrt{s_-}|e_1 f_1\rangle$ with $s_+ = 1/2 - \kappa$ and $s_- = p/2 - \delta + \kappa$. Bob's marginal gives $\kappa \ge 0$. The largest eigenvalue of σ_A is at most $s_+ + t$, so $(1+c)/2 \le 1/2 - \kappa + d + \delta$ and $\kappa \le \delta$. Lemma 6.1 and the hypothesis give $\delta \le D_{\mathrm{pur}}/p \le p/4$, hence $s_- \ge p/4 > 0$.

Let $\phi = e_0 \otimes f_1$. Bob's marginal gives $\tau_B = \kappa|f_0\rangle\langle f_0| + (t-\kappa)|f_1\rangle\langle f_1|$. With $A = \langle e_0 f_0|\tau|e_0 f_0\rangle$ and $B = \langle e_1 f_1|\tau|e_1 f_1\rangle$, the relation $\tau v = 0$ gives $s_+ A = s_- B$. Since $A \le \kappa$, the weight of τ outside ϕ is $\ell = t - \langle\phi|\tau|\phi\rangle = \kappa + B \le \kappa\lambda/s_-$. For a spectral decomposition $\tau = \sum_j \mu_j |\chi_j\rangle\langle\chi_j|$, the identity $\| |\chi_j\rangle\langle\chi_j| - |\phi\rangle\langle\phi| \|_1 = 2\sqrt{1 - |\langle\phi,\chi_j\rangle|^2}$, the triangle inequality and Cauchy–Schwarz give $\|\tau - t|\phi\rangle\langle\phi|\|_1 \le 2\sqrt{t\ell} \le 2\sqrt{t\lambda\kappa/s_-} \le 2\sqrt{\delta/p}$, using $t + \lambda = 1$; replacing t by d costs δ. Let $w = |e_0 f_0\rangle/\sqrt{2} + \sqrt{p/2}|e_1 f_1\rangle$ and $\omega = |w\rangle\langle w| + d|\phi\rangle\langle\phi|$. From $(\sqrt{x} - \sqrt{y})^2 \le |x - y|$, $\|v - w\|^2 \le \kappa + (\delta - \kappa) = \delta$, so $\| |v\rangle\langle v| - |w\rangle\langle w| \|_1 \le (\|v\| + \|w\|)\,\|v - w\| \le 2\sqrt{\delta}$. Altogether $\|\sigma - \omega\|_1 \le 2\sqrt{\delta} + 2\sqrt{\delta/p} + \delta$; then use $\delta \le D_{\mathrm{pur}}/p$, $p \le 1$ and $\sqrt{D_{\mathrm{pur}}} \le p/2$. □

Corollary 6.3 (Near saturation)**.** *Let $\varepsilon = H(c,r) - M(\rho)$ with $c < 1$, $r < 1$ and $\varepsilon \le (1-r^2)(1-c)^2/2$. Then some local-unitary image ω of ω_c satisfies $\|\sigma - \omega\|_1 \le \frac{9}{2(1-c)}\sqrt{\varepsilon/(2(1-r^2))}$.*

This follows from $D_{\mathrm{pur}} \le \varepsilon/[2(1-r^2)]$ in (10). Undoing the whitening multiplies the bound by at most $1 + r = \|F\|_{\mathrm{op}}^2$; the approximant is a filtered obese state, and exact extremality also needs the orientation condition, which the next results control.

Proposition 6.4 (Orientation deficit)**.** *For a filtered obese state with $0 < c < 1$ and $R < 1$, in the notation of* (13)*, $(H - M)(h - \lambda_-) = c(1-c)(R-z)^2$, and hence $H - M \ge c(1-c)(R-z)^2$.*

Proof. The identity is (14) with $M = q + \lambda_+$ and $H = q + h$. Since $0 \le \lambda_- \le h \le 1$, and $h - \lambda_- > 0$ (if $R > 0$ use $h - q = R$ and $\lambda_- \le q$; if $R = 0$ it equals $c(1-c)$), divide. □

Lemma 6.5 (Fixed-radius filter transport)**.** *Let ω be any density matrix with $\omega_B = \mathbb{I}/2$, let $\rho_{\boldsymbol{b}} = (\mathbb{I} \otimes F_{\boldsymbol{b}})\omega(\mathbb{I} \otimes F_{\boldsymbol{b}})$, and let $\|\boldsymbol{b}\| = \|\boldsymbol{d}\| = r < 1$. With $\gamma_r = (\sqrt{1+r} + \sqrt{1-r})^{-1}$ and $x = \|\boldsymbol{b} - \boldsymbol{d}\|$,*

$$\|\rho_{\boldsymbol{b}} - \rho_{\boldsymbol{d}}\|_1 \le 2\gamma_r x\sqrt{1 - \gamma_r^2 x^2/4} \le 2\gamma_r x,$$

with equality in the first bound when ω is maximally entangled.

Proof. At fixed radius $F_{\boldsymbol{b}} = \alpha_r \mathbb{I} + \gamma_r \boldsymbol{b}\cdot\boldsymbol{\sigma}$ with $\alpha_r = (\sqrt{1+r} + \sqrt{1-r})/2$ and $\alpha_r^2 + \gamma_r^2 r^2 = 1$. Apply both filters to a purification of ω; both vectors have norm one because $\omega_B = \mathbb{I}/2$, and their overlap is $\frac{1}{2}\operatorname{tr}(F_{\boldsymbol{b}} F_{\boldsymbol{d}}) = 1 - \gamma_r^2 x^2/2$. The trace-norm distance of two pure states is twice the square root of one minus the squared overlap, and the partial trace contracts the trace norm. For maximally entangled ω no environment is traced out. □

Theorem 6.6 (Distance to an exact extremizer)**.** *Let $0 < c < 1$, $0 \le r < 1$, $p = 1 - c$, $\varepsilon = H(c,r) - M(\rho)$ and $0 \le \varepsilon \le (1-r^2)p^2/2$. Put $d = \frac{9}{2p}\sqrt{\varepsilon/(2(1-r^2))}$, $t_0 = (1+r)d$ and $J_r =$*

$2/(\sqrt{1+r}+\sqrt{1-r})$. *Some density matrix* τ *with the same* c *and* r *and* $M(\tau) = H(c,r)$ *satisfies*

$$\|\rho - \tau\|_1 \le t_0 + J_r \sqrt{\frac{2}{1-r}} \sqrt{\frac{\varepsilon + 4t_0}{c(1-c)}}. \tag{16}$$

For fixed interior (c,r) *this is* $O(\varepsilon^{1/4})$*; when* $r = 0$ *the bound* $\|\rho - \tau\|_1 \le d$ *holds.*

Proof. Corollary 6.3 gives an obese ω with $\|\sigma - \omega\|_1 \le d$, so $\rho' = F_{\boldsymbol{b}} \omega F_{\boldsymbol{b}}$ has $\|\rho - \rho'\|_1 \le t_0$. Every CHSH operator has norm at most $2\sqrt{2}$ (its square is $4\mathbb{I} - [A, A'] \otimes [B, B']$), so $|M(\rho) - M(\rho')| \le 4 \|\rho - \rho'\|_1$ and $H - M(\rho') \le \varepsilon + 4t_0$. In the frame where $\omega = \omega_c$, Proposition 6.4 gives $c(1-c)(r^2 - b_z)^2 \le \varepsilon + 4t_0$. Let $\boldsymbol{b}^*$ have the radius and azimuth of $\boldsymbol{b}$ and $b_z^* = r^2$. With $y = \sqrt{r^2 - b_z^2}$ and $w = r\sqrt{1-r^2}$, $\|\boldsymbol{b} - \boldsymbol{b}^*\|^2 = (b_z - r^2)^2 + (y - w)^2 \le \frac{2}{1-r}(b_z - r^2)^2$, using $|b_z + r^2| \le r(1+r)$ and $y + w \ge r\sqrt{1-r^2}$. The state $\tau = F_{\boldsymbol{b}^*} \omega_c F_{\boldsymbol{b}^*}$ is an exact extremizer by Theorem 5.1; apply Lemma 6.5 with $2\gamma_r = J_r$. □

Theorem 6.7 (Rigidity with the whole centre vector fixed)**.** *Let* $\sigma_B = \mathbb{I}/2$*, let Alice's Bloch vector be* cn *with* $0 < c < 1$*,* $\|n\| = 1$*, and let* $\delta = L - \lambda_{\max}(\sigma) \le c/8$*. There is an obese state* ω *with* $\omega_A = \sigma_A$*,* $\omega_B = \mathbb{I}/2$ *and* $\|\sigma - \omega\|_1 \le \min\{2, 9\sqrt{\delta/c}\}$*. In particular, if* $D_{\mathrm{pur}} \le c(1-c)/8$*, the bound* $9\sqrt{D_{\mathrm{pur}}/[c(1-c)]}$ *holds with the same Alice centre vector.*

Proof. Set $q = c/2$, $p = 1 - c$, $\lambda = L - \delta$ and $t_1 = 1 - \lambda = q + \delta$. Write $\sigma = \lambda |\psi\rangle\langle\psi| + t_1 \eta$ with η a density matrix, let $a_\psi, b_\psi, a_\eta, b_\eta$ be the marginal Bloch vectors, $s = \|a_\psi\| = \|b_\psi\|$ and $u = \|b_\eta\|$. Bob-flatness gives $\lambda s = t_1 u$ with opposite Bob directions, and Alice's marginal gives the nonnegative decomposition

$$2\delta = t_1(1 - u) + \lambda(s - n \cdot a_\psi) + t_1(1 - n \cdot a_\eta). \tag{17}$$

In particular $t_1 u \ge q - \delta > 0$. Let e_0 point along n, f_1 along b_η/u, and $\Phi = |e_0 f_1\rangle\langle e_0 f_1|$. The two local failure projectors commute, so (17) gives $\langle e_0 f_1 | \eta | e_0 f_1 \rangle \ge 1 - \delta/t_1$ and $\|t_1 \eta - t_1 \Phi\|_1 \le 2\sqrt{t_1 \delta}$. If θ is the angle between a_ψ/s and n, the middle term gives $\sin^2(\theta/2) \le \delta/(\lambda s) \le \delta/(q - \delta)$. Rotating Alice's Schmidt axis to n about an axis perpendicular to a_ψ changes the vector by overlap $\cos(\theta/2)$, at trace-norm cost at most $2\lambda\sqrt{\delta/(q-\delta)}$. With $\kappa = t_1(1-u)/2 \in [0, \delta]$ the rotated weighted vector is $\sqrt{1/2 - \kappa}|e_0 f_0\rangle + \sqrt{p/2 - \delta + \kappa}|e_1 f_1\rangle$; compare it with $w = |e_0 f_0\rangle/\sqrt{2} + \sqrt{p/2}|e_1 f_1\rangle$ and set $\omega = w w^\dagger + q\Phi$, which is obese with the required marginals. The vector error is at most $\sqrt{\delta}$ and the rank-one error at most $(\sqrt{\lambda} + \sqrt{L})\sqrt{\delta}$; changing the weight t_1 to q costs δ. Hence $\|\sigma - \omega\|_1 \le 2\sqrt{t_1 \delta} + 2\lambda\sqrt{\delta/(q-\delta)} + (\sqrt{\lambda} + \sqrt{L})\sqrt{\delta} + \delta \le 9\sqrt{\delta/c}$ for $\delta \le c/8$. Lemma 6.1 gives the final statement. □

Consequently Theorem 6.6 holds for an exact extremizer with the same centre *vector* and Bob length under $\varepsilon \le (1 - r^2)c(1-c)/4$, with d replaced by $9\sqrt{\varepsilon/(2(1-r^2)c(1-c))}$; Bob filtering preserves the steering ellipsoid, so the orientation correction keeps the centre vector.

Lemma 6.8 (Small-centre direction correction)**.** *For* $0 \le c \le 1/2$*, any local-unitary image of* ω_c *can be moved within its orbit to any prescribed Alice centre direction at trace-norm cost at most* $3c$*.*

Proof. With Φ the projector onto $(|00\rangle + |11\rangle)/\sqrt{2}$, $\|\omega_c - \Phi\|_1 \le \sqrt{2}(1 - \sqrt{1-c}) + c/2$ in every local frame. An Alice rotation paired with $U \otimes \overline{U}$ fixes Φ, so the cost is at most $2\sqrt{2}(1 - \sqrt{1-c}) + c = (\frac{2\sqrt{2}}{1+\sqrt{1-c}} + 1)c \le 3c$. □

Corollary 6.9 (Uniform rigidity with optimal exponent)**.** *For every canonical state, including $c = 0, 1$, there is an obese state ω with the same Alice Bloch vector and*

$$\|\sigma - \omega\|_1 \le \min\{2, 9D_{\mathrm{pur}}^{1/4}\}. \tag{18}$$

No bound $KD_{\mathrm{pur}}^{\alpha}$ with $\alpha > 1/4$ and a constant K independent of c holds for all canonical states. The constant nine is not claimed optimal.

Proof. Boundary estimate. In Alice's eigenbasis write $\sigma = \left(\begin{smallmatrix} A & B \\ B^\dagger & E \end{smallmatrix}\right)$ with $A + E = \mathbb{I}/2$ and $\operatorname{tr} E = p/2$, $p = 1 - c$. Positivity gives $B = A^{1/2} K E^{1/2}$ with a contraction K, hence $\|B\|_1 \le \sqrt{\operatorname{tr} A \operatorname{tr} E}$, and with $P = |0\rangle\langle 0| \otimes \mathbb{I}/2$ we get $\|\sigma - P\|_1 \le p + \sqrt{2p - p^2}$. The aligned obese state obeys the same bound, so the distance to an obese state with the same Alice vector is at most $2p + 2\sqrt{2p - p^2} \le (2 + 2\sqrt{2})\sqrt{p}$.

Cases. Assume $0 < D_{\mathrm{pur}} \le 1/256$ and put $d = D_{\mathrm{pur}}^{1/4} \le 1/4$ (for $D_{\mathrm{pur}} > 1/256$ the bound exceeds two). If $c < 1/2$ then $p > 1/2$ and Theorem 6.2 gives distance at most $9D_{\mathrm{pur}}^{1/2} = 9d^2$ to some obese state; if moreover $c < d$, Lemma 6.8 fixes the direction at total cost $9d^2 + 3d \le 9d$, while if $d \le c < 1/2$ then $D_{\mathrm{pur}} = d^4 \le d/16 \le cp/8$ and Theorem 6.7 gives at most $9\sqrt{2}d^{3/2} \le 9d$. If $c \ge 1/2$ and $p < 2d^2$, the boundary estimate gives at most $(4 + 2\sqrt{2})d < 9d$. If $c \ge 1/2$ and $p \ge 2d^2$, then $D_{\mathrm{pur}} \le d^2/8 \le cp/8$ and $cp \ge d^2$, and Theorem 6.7 gives at most $9d$.

Optimality. For $0 < c < 1$ let

$$\sigma_c = \tfrac{1}{4}\big[\mathbb{I} \otimes \mathbb{I} + c\sigma_z \otimes \mathbb{I} + \sqrt{1 - c^2}\, \sigma_x \otimes \sigma_z\big], \tag{19}$$

an equal mixture of two pure Alice states with Bloch vectors $(\pm\sqrt{1 - c^2}, 0, c)$ paired with orthogonal Bob states. It is canonical, separable, $\sigma_c^2 = \sigma_c/2$, and $D_{\mathrm{pur}} = p^2/2$. Its correlation matrix has operator norm $\sqrt{2p - p^2}$, while every local-unitary image of ω_c has correlation operator norm $\sqrt{p}$. Testing against the norm-one observable $\sigma_x \otimes \sigma_z$ gives $\|\sigma_c - \omega\|_1 \ge \sqrt{2p - p^2} - \sqrt{p} = \sqrt{p}(\sqrt{2 - p} - 1)$ for every such ω. Divided by $D_{\mathrm{pur}}^{1/4} = 2^{-1/4}\sqrt{p}$ this tends to $2^{1/4}(\sqrt{2} - 1) > 0$ as $p \downarrow 0$, and divided by $D_{\mathrm{pur}}^{\alpha}$ with $\alpha > 1/4$ it diverges. □

The same exponent is optimal for canonical CHSH rigidity: if $\varepsilon = 2(1 - c) - M(\sigma)$ then $D_{\mathrm{pur}} \le \varepsilon/2$, so (18) gives an obese state with the same centre vector within $\min\{2, 9(\varepsilon/2)^{1/4}\}$, while (19) has $M = 1 - c^2$ and $\varepsilon = (1 - c)^2$.

7 Full-ellipsoid envelopes

The centre length alone does not determine the CHSH optimum of a fixed ellipsoid. For a fixed canonical state, (6) reduces the optimum over states with that ellipsoid and Bob length r to the choice of the filter direction; right orthogonal changes of C preserve the range of $C\boldsymbol{b}$, so the optimum depends only on the ellipsoid. We solve this for three classes.

7.1 Translated spheres

Let $Q = s^2\mathbb{I}$ and $\boldsymbol{c} = ce_z$ with $s, c \ge 0$ and $s + c \le 1$; the representative $C = s\,\mathrm{diag}(1, -1, 1)$ is physical.

Theorem 7.1 (Translated-sphere envelope)**.** *For $0 \le r < 1$ the maximum of M over states with this ellipsoid and Bob length r is $2s^2 + (c^2 - s^2)r^2 + 2scr$; for $s, c, r > 0$ it is attained exactly when $C\boldsymbol{b} = sre_z$. For $s > 0$ its supremum over $r < 1$ is $2s^2 + s^2c^2/(s^2 - c^2)$ at $r_* = sc/(s^2 - c^2)$ if $sc < s^2 - c^2$, and otherwise $(s + c)^2$, approached as $r \to 1$.*

Proof. For $sr > 0$ let $n = C\boldsymbol{b}/(sr)$. By (6), $TT^{\mathsf{T}} = q\mathbb{I} + K$ with $q = s^2(1-r^2)$ and $K = s^2r^2nn^{\mathsf{T}} + scr(ne_z^{\mathsf{T}} + e_zn^{\mathsf{T}}) + c^2r^2e_ze_z^{\mathsf{T}}$, which is supported on $\mathrm{span}\{n, e_z\}$ with nonpositive determinant there, so $M = 2q + \lambda_{\max}(K)$. For every unit x, $x^{\mathsf{T}}Kx \le s^2r^2 + 2scr + c^2r^2$, with equality when $n = x = e_z$, and for $s, c, r > 0$ equality in the cross term forces this alignment. Then maximize the quadratic in r; its vertex lies in $[0, 1)$ exactly when $sc < s^2 - c^2$. □

For $s = 2/3$, $c = 1/3$ the canonical state has $M = 8/9 < 1$, while $\rho = \frac{1}{18}\begin{pmatrix} 15 & 0 & 0 & 2\sqrt{5} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2\sqrt{5} & 0 & 0 & 2 \end{pmatrix}$ has the same ellipsoid and $M = 28/27 > 1$: whitening need not maximize CHSH within a fixed ellipsoid. The loss-channel boundary $c = 1 - s$ overlaps Appendix B of [3].

7.2 Axial ellipsoids

Take $\boldsymbol{c} = ce_z$ and $C = \mathrm{diag}(s, -s, t)$ with $s, t, c \ge 0$, and put $S = s^2$, $T = t^2$, $D = S - T$. The canonical density matrix $\frac{1}{4}\begin{pmatrix} 1+c+t & 0 & 0 & 2s \\ 0 & 1+c-t & 0 & 0 \\ 0 & 0 & 1-c-t & 0 \\ 2s & 0 & 0 & 1-c+t \end{pmatrix}$ is positive exactly when

$$c + t \le 1, \qquad 4s^2 + c^2 \le (1+t)^2. \tag{20}$$

The representative $\mathrm{diag}(s, s, t)$ needs $4s^2 + c^2 \le (1-t)^2$, so whenever it is physical so is the displayed one.

Theorem 7.2 (Axial envelope at fixed Bob length)**.** *For $0 \le r < 1$,*

$$\max_{\|\boldsymbol{b}\|=r} M = \begin{cases} 2S + Sr^2(c^2/D - 1), & D > 0 \text{ and } ctr \le D, \\ S + T + 2ctr + (c^2 - S)r^2, & D \le 0 \text{ or } ctr \ge D, \end{cases} \tag{21}$$

the two expressions agreeing at $ctr = D > 0$. For $r > 0$ the optimizing filter directions (any azimuth when a latitude is given) are: $b_z = ctr^2/D$ if $D > 0$ and $ctr \le D$; $\boldsymbol{b} = re_z$ if $D > 0$ and $ctr \ge D$, if $D = 0 < c$ with $s = t > 0$, or if $D < 0 < c$; $\boldsymbol{b} = \pm re_z$ if $D < 0 = c$; every direction if $D = c = 0$ or $s = t = 0$.

Proof. By symmetry $\boldsymbol{b} = (\sqrt{R - z^2}, 0, z)$ with $R = r^2$, $|z| \le r$. The Gram matrix has the eigenvalue $q = S(1 - R)$ and the block

$$B(z) = \begin{pmatrix} S(1 - z^2) & s\sqrt{R - z^2}(c + tz) \\ s\sqrt{R - z^2}(c + tz) & (1 - R)(T - c^2) + (c + tz)^2 \end{pmatrix},$$

with $\det(q\mathbb{I} - B) = -S(R - z^2)(1 - R)(D + c^2)$. If $D + c^2 \ge 0$ then $\lambda_- \le q \le \lambda_+$ and $M = q + \lambda_+$; if $D + c^2 < 0$ the first diagonal entry is at least q, both block eigenvalues are at least q and $M = \mathrm{tr}\, B$ (for $S = 0$ the block is diagonal).

Interior branch ($D > 0$, $ctr \le D$). With $h_i = S + Sc^2R/D$ and $z_i = ctR/D \in [0, r]$,

$$(h_i\mathbb{I} - B)_{11} = Sz^2 + Sc^2R/D, \qquad \det(h_i\mathbb{I} - B) = S(D + c^2)(z - z_i)^2.$$

The first entry is positive unless $z = cr = 0$, when the off-diagonal entry vanishes and the other gap is $S - (1 - R)T \ge D > 0$. So $h_i\mathbb{I} - B \succeq 0$, with singularity exactly at $z = z_i$.

Boundary branch. With $h_b = (t + cr)^2$, $A_b = h_bD + qc^2$ and $L_b = (c + rt)(t + cr)(ctr - D)$,

$$(h_b\mathbb{I} - B)_{11} = -D + 2ctr + c^2R + Sz^2, \qquad \det(h_b\mathbb{I} - B) = A_b(r - z)^2 + 2L_b(r - z),$$
$$A_br - ct(h_b - q) = (c + rt)(t + cr)(D - ctr).$$

If $D \geq 0$ and $ctr \geq D$ the determinant terms are nonnegative, and for $D > 0$ (or $D = 0$ with $s = t$, $c, r > 0$, where $q > 0$) equality holds only at $z = r$. In the prolate interlacing case $D < 0 \leq D + c^2$ one has $c, t > 0$, the first gap is positive, and with $v = r - z \in [0, 2r]$ the determinant is $v(A_b v + 2L_b)$, whose affine factor is $2L_b > 0$ at $v = 0$ and $2ct(h_b - q) > 0$ at $v = 2r$; so the determinant is nonnegative, vanishing only at $z = r$. Finally, if $D + c^2 < 0$, $q + (t + cr)^2 - M = (r - z)[(T - S)(r + z) + 2ct] \geq 0$, vanishing only at $z = r$ when $c > 0$ and at both poles when $c = 0$. The degenerate spheres and points follow from the block directly. □

Optimizing the Bob length. Let $\mathcal{M} = \sup_{r<1} \max_{\|\boldsymbol{b}\|=r} M$ and $U = S - c^2$. If $D > 0$ and $c^2 < D$ then $\mathcal{M} = 2S$, attained only at $r = 0$; if $D > 0$ and $c^2 = D$ the same value is attained at every radius with $ctr \leq D$. Otherwise ($D \leq 0$ or $c^2 > D$): if $U > ct$ then $\mathcal{M} = S + ST/U$ at the unique radius $r_* = ct/U$, with the aligned direction; if $U \leq ct$ then $\mathcal{M} = S + Sc^2/D$ when $D > 0$ and $ct \leq D$, and $\mathcal{M} = (t + c)^2$ when $D \leq 0$ or $ct \geq D$, approached as $r \uparrow 1$ and not attained except for a centred line or a point. Indeed the two branches $F_i(r) = 2S + S(c^2 - D)r^2/D$ and $F_b(r) = S + T + 2ctr - Ur^2$ agree to first order at $r = D/(ct)$, and when F_i increases the vertex of F_b lies beyond the transition, since $ct/U - D/(ct) = S(c^2 - D)/(ctU) > 0$. The boundary value is the largest squared norm of a point of the ellipsoid, at most one by physicality.

Corollary 7.3 (Axial CHSH-breaking criterion)**.** *The physical qubit channel $x \mapsto ce_z + \mathrm{diag}(s, s, t)x$ is CHSH-breaking, i.e. no two-qubit input yields $\beta > 2$ after the channel acts on one qubit, if and only if*

$$2S \leq 1 \quad \textit{and} \quad \big[S \leq c^2 + ct \ \textit{or}\ (1 - S)(S - c^2) \geq ST\big]. \tag{22}$$

Proof. The canonical state above is the channel's Choi state. Every pure input is a maximally entangled state filtered on the untouched qubit, and the filter commutes with the channel, so the outputs of pure inputs with invertible filters form the orbit optimized above; product inputs (singular filters) give products, which are CHSH-local, and convexity of β reduces mixed inputs to pure ones. The supremum is $2S$, the vertex value $S + ST/(S - c^2)$ when $S - c^2 > ct$, or a boundary value at most one, and the first two are at most one exactly under (22). □

The criterion concerns arbitrary inputs with one channel use and no further filtering; filtering on both sides, as in Pal and Ghosh [6], and Choi-state nonlocality, as in Rutkowski and Siudzińska [7], are different questions. The unital and amplitude-damping cases are in [6, 3].

7.3 An exact witness requiring a tilted filter

Take $c = 1/5$, $t = 3/5$, $S = 3/5$, $r = 1/2$; (20) holds strictly and $ctr = 3/50 < D = 6/25$. The interior branch gives $M = 43/40 > 1$ at the canonical-frame filter $\boldsymbol{b} = (\sqrt{15}/8, 0, 1/8)$, while the aligned filter at the same radius gives $M = 47/50 < 1$. In a rotated Bob basis an attaining state is

$$\rho_{\mathrm{ax}} = \frac{1}{160}\begin{pmatrix} 81 & -9\sqrt{5} & 45 & 15\sqrt{5} \\ -9\sqrt{5} & 21 & -9\sqrt{5} & -15 \\ 45 & -9\sqrt{5} & 39 & 9\sqrt{5} \\ 15\sqrt{5} & -15 & 9\sqrt{5} & 19 \end{pmatrix}, \tag{23}$$

which is positive definite: $\rho_{\mathrm{ax}} = LDL^{\mathsf{T}}$ with $D = \mathrm{diag}(81/160, 1/10, 9/160, 1/90)$ and unit lower-triangular L with rows (1), $(-\sqrt{5}/9, 1)$, $(5/9, -\sqrt{5}/4, 1)$, $(5\sqrt{5}/27, -5/12, -\sqrt{5}/9, 1)$. Its Bloch data are $\boldsymbol{a} = (3/8, 0, 11/40)$, $\boldsymbol{b} = (0, 0, 1/2)$ and $T = \begin{pmatrix} 3\sqrt{5}/40 & 0 & 3/4 \\ 0 & -3\sqrt{5}/10 & 0 \\ -9\sqrt{5}/40 & 0 & 1/4 \end{pmatrix}$, so $\boldsymbol{c} = (0, 0, 1/5)$,

$Q = \mathrm{diag}(3/5, 3/5, 9/25)$ up to rotation and $T^{\mathsf{T}}T = \mathrm{diag}(9/32, 9/20, 5/8)$. The unit settings $\alpha = (3, 0, 1)/\sqrt{10}$, $\alpha' = (0, -1, 0)$, $\beta = (0, \sqrt{18}, 5)/\sqrt{43}$, $\beta' = (0, -\sqrt{18}, 5)/\sqrt{43}$ give $\mathrm{tr}(\rho_{\mathrm{ax}}B) = 2\sqrt{43/40}$. A second example, $S = 1/2$, $t = 2/3$, $c = 1/3$, has canonical value $M = 1$ and global maximum $M = 15/14$ at the aligned filter with $r = 4/7$.

7.4 Centred ellipsoids

Theorem 7.4 (Centred envelope)**.** *Let $\boldsymbol{c} = 0$ and let $\ell_1 \geq \ell_2 \geq \ell_3 \geq 0$ be the eigenvalues of $Q = CC^{\mathsf{T}}$. For $0 \leq r < 1$, $\max_{\|\boldsymbol{b}\|=r} M = \ell_1 + (1 - r^2)\ell_2$. For $r > 0$ equality holds exactly when $\boldsymbol{b}/r$ lies in the top eigenspace of $C^{\mathsf{T}}C$ (all directions if $Q = 0$).*

Proof. With $\boldsymbol{b} = rn$, $G = TT^{\mathsf{T}} = (1 - r^2)Q + r^2 Cnn^{\mathsf{T}}C^{\mathsf{T}}$. For every rank-two orthogonal projection P, $\mathrm{tr}(PG) \leq (1 - r^2)(\ell_1 + \ell_2) + r^2 \|Cn\|^2 \leq (1 - r^2)(\ell_1 + \ell_2) + r^2\ell_1$. Choosing n in the top right eigenspace and P containing Cn and a second leading left eigenvector attains both bounds; for $0 < r < 1$ equality forces $\|Cn\|^2 = \ell_1$. □

Over all radii the supremum is the canonical value $\ell_1 + \ell_2$. The corresponding CHSH-breaking criterion for unital qubit channels is prior work [6, 3]; the fixed-radius value and equality cases are the finer statement here.

8 A printed optimizer in Zhang et al.

Appendix B of [3] (arXiv v2 and the published version) prints an optimizing vector with a repeated R_{31} and last coordinate $R_{33} - (1 - \alpha^2)$, and its prose states $R_{33} = 1 - \alpha^2$. Re-expanding its preceding correlation matrix gives, for $r = \sqrt{1 - \alpha^2}$, $H' = -p(1-p)aa^{\mathsf{T}} + r^2 e_z e_z^{\mathsf{T}}$ with $a = (\alpha R_{31}, \alpha R_{32}, R_{33} - r)$, so the optimizing condition is $R_{33} = r$, which is $b_z = \|\boldsymbol{b}\|^2$ in filter coordinates. For $p = 1/2$, $r = 1/2$, $\alpha = \sqrt{3}/2$ and a y rotation with $R_{33} = 1/4$, the correlation matrix is $T = \begin{pmatrix} \sqrt{6}/16 & 0 & -3\sqrt{5}/16 \\ 0 & -\sqrt{6}/4 & 0 \\ \sqrt{30}/8 & 0 & 3/8 \end{pmatrix}$ and $M = (399 + 9\sqrt{145})/512 < 1$, strictly, since $113^2 - 81 \cdot 145 = 1024 > 0$; the printed direction does not attain the maximum, which the corrected direction does. The maximal value and the CHSH-breaking threshold of [3] are correct.

9 Verification

9.1 Lean

The package contains a standalone Lean project (Lean 4.19.0, Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`) of fifteen modules and a whole-project axiom audit. Its definitions are explicit: complex 4×4 matrices; density matrices as Hermitian, trace one and $\mathrm{Re}\langle w, \rho w\rangle \geq 0$ for *every* complex vector w; the standard Pauli matrices and the tensor product in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; the Bell operator above with four real unit vectors; the modulus $\sqrt{\mathrm{normSq}}$ of the complex expectation $\mathrm{tr}(\rho B)$; and β as the real supremum of these values.

- `CertifiedCHSH.physical_joint_chsh_attainment`: for every $0 \leq c \leq 1$ and $r^2 < 1$ the matrix $\rho^{\sharp}_{c,r}$ of Proposition 5.2 is a density matrix, the centre formula (5) applied to its Bloch data gives $(0, 0, c)$, Bob's length is r, and $\beta(\rho^{\sharp}_{c,r}) = 2\sqrt{1 + (1 - 2c)(1 - r^2)}$. The Bloch data are tied to the matrix by the readout theorems `CHSHPhysical.alice_readout`, `bob_readout` and `correlation_readout`. The upper bound covers every setting through an elementary

orthogonal-plane certificate, and explicit unit settings attain it; the Horodecki formula is not assumed.

- `CHSHAxialWitness.beta_exact` and related theorems: the matrix (23) is a density matrix with the stated Pauli data and $\beta = 2\sqrt{43/40}$.
- `BoundarySharpness`: the family (19) is a canonical density matrix with $\rho^2 = \rho/2$, purity $1/2$, the stated Pauli data, exact β and canonical CHSH gap $(1-c)^2$.
- `AxialCertificates.axial_exact_maximum`: for $S, c, t, r \geq 0$, $r^2 \leq 1$ and $|z| \leq r$, the largest sum of two of the three roots of the axial Gram matrix (the explicit radical roots of the block and q) is at most the envelope (21), with an attaining z; neither physicality nor the inequality is assumed. Further modules check the obese block algebra, the orientation correction, spectral purity inequalities and the countercertificate of Section 8.

A fresh replay compiled all modules with warnings as errors and audited all 839 declarations owned by the fifteen modules: the 543 declarations other than code-generation auxiliaries are not axioms and depend only on `propext`, `Classical.choice` and `Quot.sound`; the 296 auxiliaries stored by the Lean 4.19 code generator are recorded separately, and one of them refers to the generator's placeholder `lcProof`. The sources contain no `sorry`, added axiom or `native_decide`.

Not formalized: the arbitrary-state bound (2) (it is prior work); the general filter-to-Gram correspondence (6) and the Horodecki identification for arbitrary states, which connect the theorem `axial_exact_maximum` to Theorem 7.2; the equality classification; the rigidity theorems and the lower bound against arbitrary local-unitary images; and the operational meaning of the centre.

9.2 Exact and numerical checks

An exact SymPy program checks 46 identities: the Bloch data, centre, Gram characteristic polynomial and $M = H$ for $\rho^\sharp_{c,r}$ symbolically; the block identities (14) and their agreement with (6); the Gram formula on an explicitly filtered obese state; all axial identities in the proof of Theorem 7.2; the axial witness (positivity, readouts, whitening, shape and Bell value); the sharpness family; the countercertificate of Section 8; and the examples ρ_* and the translated sphere. A seeded NumPy program searches for counterexamples: 10^4 random states of ranks one to four against (2) and (10); the equality orbit and Proposition 6.4; brute-force filter directions for 300 random axial shapes against (21); random centred ellipsoids against Theorem 7.4; the channel criterion against a radius grid; and the sharpness lower bound. No violation beyond rounding ($\sim 10^{-15}$) was found. These searches are falsification attempts, not proofs.

10 Limitations

The exact optimum for an unrestricted displaced triaxial ellipsoid remains open; (6) reduces it to an optimization over Bob's Bloch ball for each measurement plane, which is not solved here. The shape-gap bound (10) is not sharp for general ellipsoids. The rigidity constants are not claimed sharp; the uniform exponent is. The full text of Xing and Yang [8], on volume-based bounds, was not available, so that paper is not excluded as an overlapping source.

11 Research process and AI disclosure

This is fully autonomous agentic research. Autonomous AI research agents produced the mathematics, the Lean development, the computations, an audit of the public Omega proof and an earlier version

of this text in earlier sessions; their records do not name the system used, and the reviews in those records were AI-assisted reviews within the same workflow. Opus 5.5 by Anthropic used with Claude Code assessed the package for this release: it compared the results with Conjecture 1, the Omega dossier and source, and the 2020 paper, and determined what is new; read the central proofs; wrote and ran its own exact and numerical check programs; replayed the Lean project with a whole-project axiom audit; rewrote this manuscript; and integrated the package into the repository. The audit and replay records of the third-party Omega project are not included. Dev D. Goyal is responsible for publication. The work has not been externally peer reviewed. The formal scope and remaining limits are stated in Sections 9 and 10.

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